is in this sense that the existence of unemployment is essential for the reproduction of capitalist production relations. Unlimited continuance of the upward cumulative disequilibrium process violates this condition.

4.3.1.4 Inevitability of Economic Crisis

In a capitalist system, the upward disequilibrium process that once emerged has a cumulative nature. However, if this cumulative process proceeds unlimitedly, capitalist production relations become impossible to reproduce themselves. Therefore, as long as a capitalist system is to continue, the upward disequilibrium process must be reversed by whatever trigger. For that reason, an economic crisis is inevitable.

As one can understand from the above explanation, an economic crisis is inevitable because the disequilibrium cumulates inevitably, and the cumulativeness of the disequilibrium is inevitable because a capitalist system is a class society in which commodity production is dominant. In other words, it is the fundamental character of a capitalist system on which the foundation of the inevitability of economic crisis is grounded.

4.3.2 Triggers of Economic Crisis

We showed the following in the previous section: In a capitalist economy, cumulative disequilibrium is inevitable. Unless this cumulative disequilibrium process is reversed, it will become impossible for a capitalist system to reproduce and sustain itself. Therefore, *as long as* a capitalist system continues, it is vital for the disequilibrium cumulative process to reverse its direction. It is the economic crisis which reverses the upward disequilibrium cumulative process to a downward one. For that reason, an economic crisis is *inevitable* as long as a capitalist system continues.

However, in Sect. 4.3.1, we did not discuss what kind of *triggers* cause an economic crisis. We must not confuse the question as to why an economic crisis is inevitable in a capitalist system with what triggers an economic crisis. Economic crises can be caused by various kinds of triggers. There is *no* inevitability as to what kind of trigger it ought to be. However, whatever the trigger is, the upward disequilibrium cumulative process must be reversed and, once reversed, there has to arise a downward disequilibrium cumulative process. This is a precondition.

The relationship between the two questions above is easy to understand as we think about the following analogy: When thinking of human death, let us consider the following two related, but different questions. One is whether or not human death is inevitable; the other one is what kind of triggers bring about death to humans. The answers to those two questions are entirely different. The former one can be answered by humans' characteristics as an organism or moreover, by some propositions of thermodynamics. The latter one can be answered by listing all the triggers of death, such as disasters, illness, or suicide. In the former sense, the inevitability of human death can be proven, which the author believes is possible; but in the latter sense, it is *impossible* to demonstrate its inevitability *generally* and *abstractly*. We cannot assure that human *always* die from cancer or *always* die in a car accident. No matter what kind of trigger is brought in, it is incorrect to claim that any human always dies from that certain trigger. As for the latter question, we can only discuss either (1) what kind of triggers' *probability* is high in general, or (2) what kind of triggers are inevitable for humans in the specific time, place and society.

We can say similar things about economic crises as well. On the general and abstract presupposition that it is a capitalist society, whatever the trigger is, we can make a case for the inevitability of economic crisis. However, trying to demonstrate that every economic crisis is always caused by a specific trigger, e.g., overproduction in the consumption goods sector or imbalance between supply and demand in the labor market, from just a general and abstract characteristic of a capitalist system, is flawed. What we can say about the triggers of an economic crisis from a general and abstract definition of a capitalist system is to show what kind of triggers have a high degree of probability in causing an economic crisis. That is all we can discuss regarding the triggers of an economic crisis as this book covers the capitalist system in general.

For instance, if the capitalist society in Japan enters an economic crisis and we try to find out its trigger, though it is an important task, we should not (and cannot) answer it just from the definition of a capitalist system in general. We ought to infer a specific trigger based on detailed investigation and research of concrete international and domestic conditions. This, however, is not the goal of this book. Nevertheless, showing what can be listed as primary triggers when premising a capitalist system, in general, will help more in-depth research.

4.3.2.1 Overproduction in Consumption Goods Sector

As already examined in Sects. 2.4.3.3 and 4.2.1, an extremely high increase in accumulation demand, which characterizes the upward cumulative disequilibrium process = upward process of business cycle, causes an increase of production goods price compared to consumption goods price, and it *may* suppress the profit rate of the consumption goods sector. On one hand, an increase in accumulation demand lowers the real wage rate, and it *may* also suppress the laborers' consumption demands. Thus, an increase in accumulation demand squeezes the demand for consumption goods and may lower profit rates and decrease production. When such situations are generated, the accumulation demand in the consumption goods sector decreases. *In such a case*, capitalists in the production goods sector become pessimistic and decrease accumulation demand so that the whole accumulation demand in the society decreases, and as a result, a downward cumulative process begins; the decrease in aggregate demand leads to the contraction of production before it decreases employment.

In order for an economic crisis to be triggered as above, the following conditions are necessary:

(a) When accumulation demand, which is mainly the demand for production goods, is durable, the production goods price increases more rapidly than the consumption goods price.
(b) This relative increase of production goods price has several negative impacts on the profit rate of the consumption goods sector so that it cannot be offset with a price increase in the consumption goods sector faster than the money wage rate.
(c) A decrease in the profit rates and a reduction in production in the consumption goods sector decrease accumulation demand in the consumption goods sector, and the latter has a negative impact big enough to damp accumulation decisions of capitalists in the production goods sector.

Whether or not those conditions are satisfied depends on various factors, such as the pricing policy of capitalists in the production goods sector, weights of production goods input in consumption goods production, movements of the money wage rate, and weight of the consumption goods sector in accumulation decisions by capitalists in the production goods sector, among others. Therefore, when the combination of these factors does not satisfy the above conditions, an increase in accumulation demand may not cause a decrease in profit rates and declining production in the consumption goods sector. Even if they occur in the consumption goods sector, a decline of the whole society may not be caused by the "run-away" of accumulation demand in the production goods sector.

4.3.2.2 Underproduction in Production Goods Sector

Even if capitalists' accumulation demand is high, when the maximum of production capacity in the production goods sector is reached, capitalists cannot get as many additional production goods as they demand. A bottleneck arises in the production goods sector. Consequently, capitalists' plans for production expansion in the next period and after will be miscarried, and this may make capitalists' accumulation demand decrease. This is a possibility. However, when capitalists' desire to accumulate is extremely strong, it may not be foiled for this reason. They may expect that production in the production goods sector will be expanded in response to the remaining demand that is left unfulfilled in the production goods sector, and they may carry through their plans for capital accumulation without decreasing the rates. Another possibility is that if the sharp increase in the production goods price occurs as a result of extreme excess demand in the production goods sector, it will lead to a decrease in the profit rate in the consumption goods sector and then to a reduction of production in this sector. This acts as a trigger, as explained in 3.2.1.

4.3.2.3 Limits of Availability of Labor Power

Along with the accelerated increase in accumulation demand, the levels of aggregate demand and aggregate production also increase at a correspondingly rapid pace, and as the demand for labor increases, unemployment decreases. Consequently, supply and demand become tight in the labor market and it may reach a point where it is difficult to obtain labor power, presuming that economic crisis does not occur by some triggers before that. As a result, the *money* wage rate will increase. However, as already examined in detail in Sect. 4.2.2, it is not correct to argue that it causes the *real* wage rate to increase and the exploitation rate to fall, and that these then cause a decrease in the accumulation demand so that an economic crisis occurs. Yet, the difficulty of obtaining labor power can be *one* of the triggers for an economic crisis, not through this path of an increase in the money wage rate leading to a decrease in the exploitation rate and then to a decrease in accumulation demand, but through totally different paths. This will be explained in the following:

(a) Similar to the case of a bottleneck in the production goods sector, the difficulty of obtaining labor power may derail capitalists' plans for production expansion in the next period and after, and as it forces capitalists to adjust their accumulation demand, it may lead to a decrease in accumulation demand, which may become a trigger for a downward reversal. However, it does not mean this always happens. Capitalists will make an effort to hire the labor away from other capitalists, not being held back by the general tightness of demand and supply of labor power. In addition, they may try to introduce so-called labor-saving new technologies that are superior in labor productivity, and therefore, they may increase accumulation demand for equipment and others even further.
(b) When facing the difficulty of obtaining labor power, a fierce battle for labor occurs among capitalists. In the booming phase, in which accumulation demand is increasing at an accelerated rate, capitalists in the consumption goods sector may lose this battle. Then, there arises idle production capacity in the consumption goods sector, while accumulation demand in this sector decreases. This may become a trigger for an overall decrease in accumulation demand, including the production goods sector, so that an economic crisis may occur.

The following reasons explain why the capitalists in the consumption goods sector lose the battle for labor power. The main instrument of this battle is the increase in the money wage rate. Capitalists who cannot match their rival's high money wage rate will lose labor power and cannot operate their equipment. The ability to raise the money wage rate depends on the increase rate of the price of the commodity they produce. During the booming phase, accumulation demand serves as the driving force and increases the aggregate demand so that the prices of commodities increase. Among them, the increasing rate of the price of production goods becomes higher than the others. It is because accumulation demand is mostly demand for production goods and the increase in the demand for consumption goods is induced by the former. The price of consumption goods may increase more than money wage rate, but the price of production goods becomes higher than that, so that, as mentioned in

Sect. 4.3.2.1, the profit rate in consumption goods sector worsens, and they lose the battle. Note that this decrease in the profit rate in consumption goods sector is not caused by a decrease in the exploitation rate, but by a relative increase in the price of production goods.

The above reason explains why the unavailability of labor power may become a trigger for economic crisis through the path of (b), but it does *not necessarily* cause an economic crisis even when the profit rate decreases and idle equipment arises in the consumption goods sector through that path. It is because the capitalists in consumption goods sector, who fall in this "trap", may try hard to get out of it and make accumulation demand to introduce new technologies. Even if the accumulation demand in the consumption goods sector decreases due to the "trap", capitalists in the production goods sector, who enjoy the excess demand, may implement more accumulation demand. In that case, the labor power will be increasingly hired away from the consumption goods sector, making it shrink, and upward the cumulative process will continue.

4.3.2.4 Lower Limit of Real Wage

As emphasized in Sect. 2.4, as long as we assume that production technology is fixed, when accumulation demand increases, the real wage rate *decreases*. Even though new technology that improve labor productivity is introduced, if the increase rate of accumulation demand is faster than productivity growth, the real wage rate will decrease after all. The real wage rate decreases either through a decrease in life materials that a laborer receive in a day, or through extension of labor hours per day. Against them, there is a limit that cannot be crossed (cf. Sect. 2.4.1)—it is the social and physiological lower limit for labor reproduction. If the real wage rate falls below this limit, one of the following will necessarily happen: (1) withering and weakening of the labor power; (2) disappearance of wage laborers in the labor market as they would not see any benefits in selling the labor power and getting employed; (3) Revolt of the wage labor class.

Assumed if the accumulation demand accelerated and the real wage rate reached close to this lower limit, the increase rate of the money wage rate would be comparatively high. It should be noted that the upward cumulative process does not necessarily go on until the real wage rate decreases to that lower limit; before reaching it, some other factors may trigger a downward momentum. However, accumulation demand is so strong that prices of commodities hike more. A severe spiral arises, money becomes useless for capitalist's calculation regarding future profits, and capitalists' accumulation demand may decrease.[16]

However, that is not the sole possibility. Even if the real wage rate moves close to the lower limit, the capitalist class may use government power to keep the money wage rate low by compulsion and make the labor class accept it. Then, as mentioned above, the weakening of the labor power will happen and laborers will lose the

[16] Joan Robinson calls it "inflation-barrier". See Robinson [52, Chap. 5, pp. 48–50].